COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff fulfil; drivers deliver; administrators report. Deployment is local-only: one host, all four roles in the same app (§8). Stock checks, confirmation transaction (§5) and role gates (FR-07) guard integrity. All stored data is course demo seed data; no real personal data is processed.

2. Development process

An increment is a unit of work closed by a review or a recorded test run; triggers may name the state that demanded the work (checks outstanding, review readiness); a date range spans multi-day work.

Table A. Increments

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Coursework kickoff	Scope, responsibilities and contracts — agreed endpoint and data shapes — recorded (increment register)
17 Oct 2026 review	Order and stock functions ready for review	Defects recorded in the increment register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Review found a confirmation endpoint that accepted repeat submits and a missing ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Order, stock, driver and admin functions complete	Application integrated; all four role workflows runnable end to end
Driver and privacy, 2 Nov 2026	Driver assignment tests and data-minimisation checks outstanding	Driver assignment and data-minimisation checks pass (tests)
Diagram and traceability review, 10–11 Nov 2026	Review found diagrams and traceability diverging from the implementation	Diagrams and traceability aligned (diagram review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date CSV export in three in-app interface interactions; the 11 Nov regression exercised all four role workflows (browser checks)
CI health tests, 11–12 Nov 2026	The record of CI-run outcomes (observations) had no regression test	Two regression tests added; suite reached 38 (unit.log)

3. Requirements

Functional requirements

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test
FR-02	enforce lifecycle and stock reservation/release	Test (cancel tests)
FR-03	number orders and shipments and show owner tracking	Test
FR-04	show inventory, stock-movement ledger and low-stock alerts; replenishment is recorded as a stock addition by staff	Test
FR-05	record notifications for order events and allow reading	Test
FR-06	provide admin aggregates and CSV export	Test
FR-07	authenticate and enforce role/object access	Test
FR-08	assign shipped orders and record delivery	Test

- NFR-01 performance — API p95 ≤2 s, first contentful paint p95 ≤3 s (budgets set in the srs before measurement, unchanged) (measurement, §6). - NFR-02 CI health — green build, test and browser runs, recorded per CI run (observations). - NFR-03 least privilege — role gates on mutating endpoints; the delivery address is released only to the assigned driver (privacy tests). - NFR-04 interaction budget — at most three in-app interface interactions per core operation (the eleven listed in §8); one interaction is a fill, selection or click; login/navigation are setup (walkthrough). - NFR-05 process quality — green CI per increment (observations).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	scheduling plan	register
RK-02 contract rework	Constant tests	transition tests
RK-03 test-week capacity	Traceability	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	Seed tests	seed regression test
RK-06 coursework assignments and test-week squeeze	Weekly re-planning	risk plan
RK-07 stack unfamiliarity	Setup guide	setup guide
RK-08 report and presentation-video deadlines	Weekly re-planning	risk plan

4. System models

The four layers — pages, routes, handlers, Prisma/SQLite persistence (ORM over the SQLite database) — separate interface, application and data concerns. Figures 1–5 answer, in order: what each role can do; how orders transition and stock changes; how entities relate; how the confirmation transaction proceeds; how the layers connect.

Each item references one product; each order has one or more items; each order has at most one shipment; products accumulate ledger entries; notifications relate to at most one order.

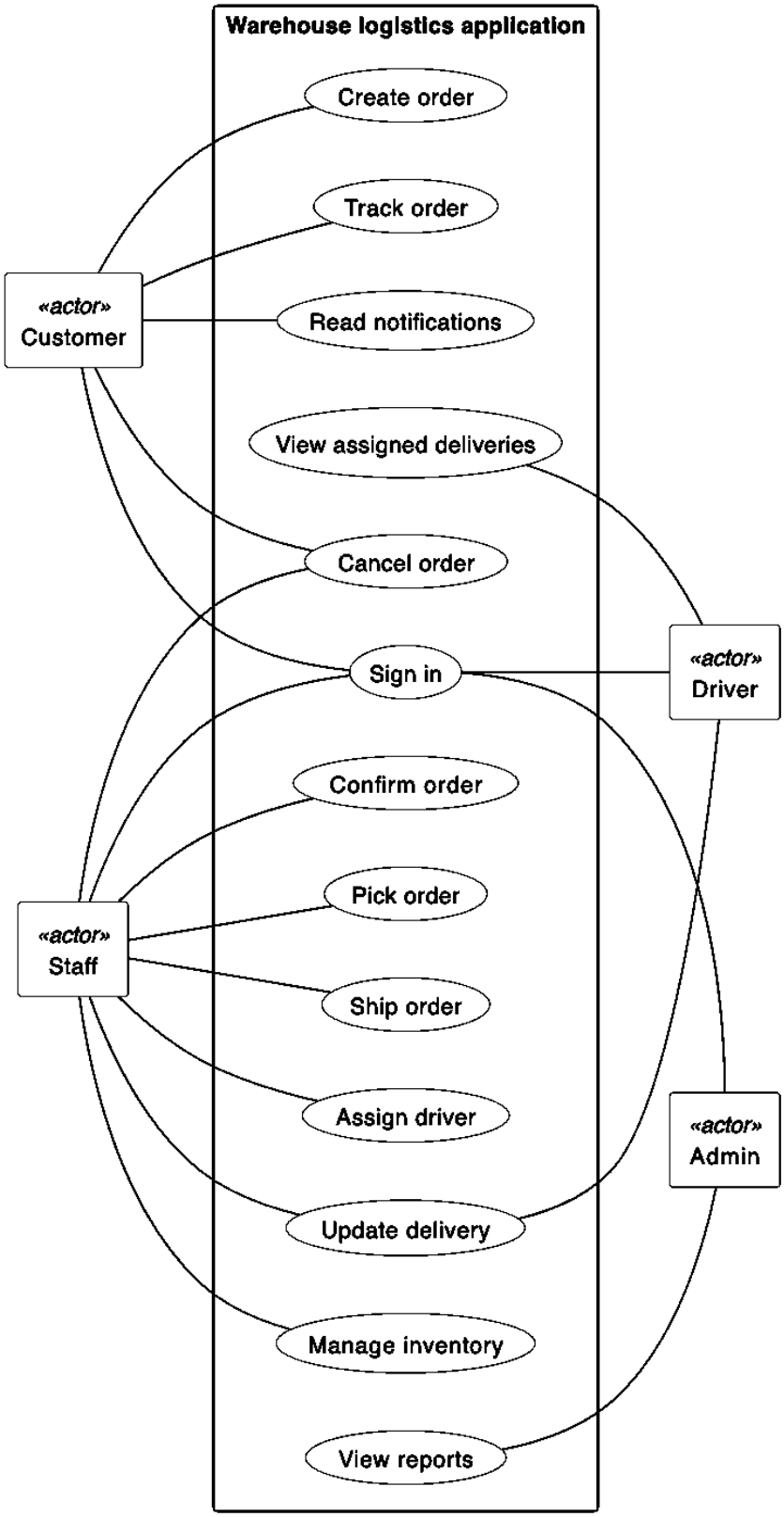


Figure 1. Role functions: thirteen use cases grouped under four role actors. Source.

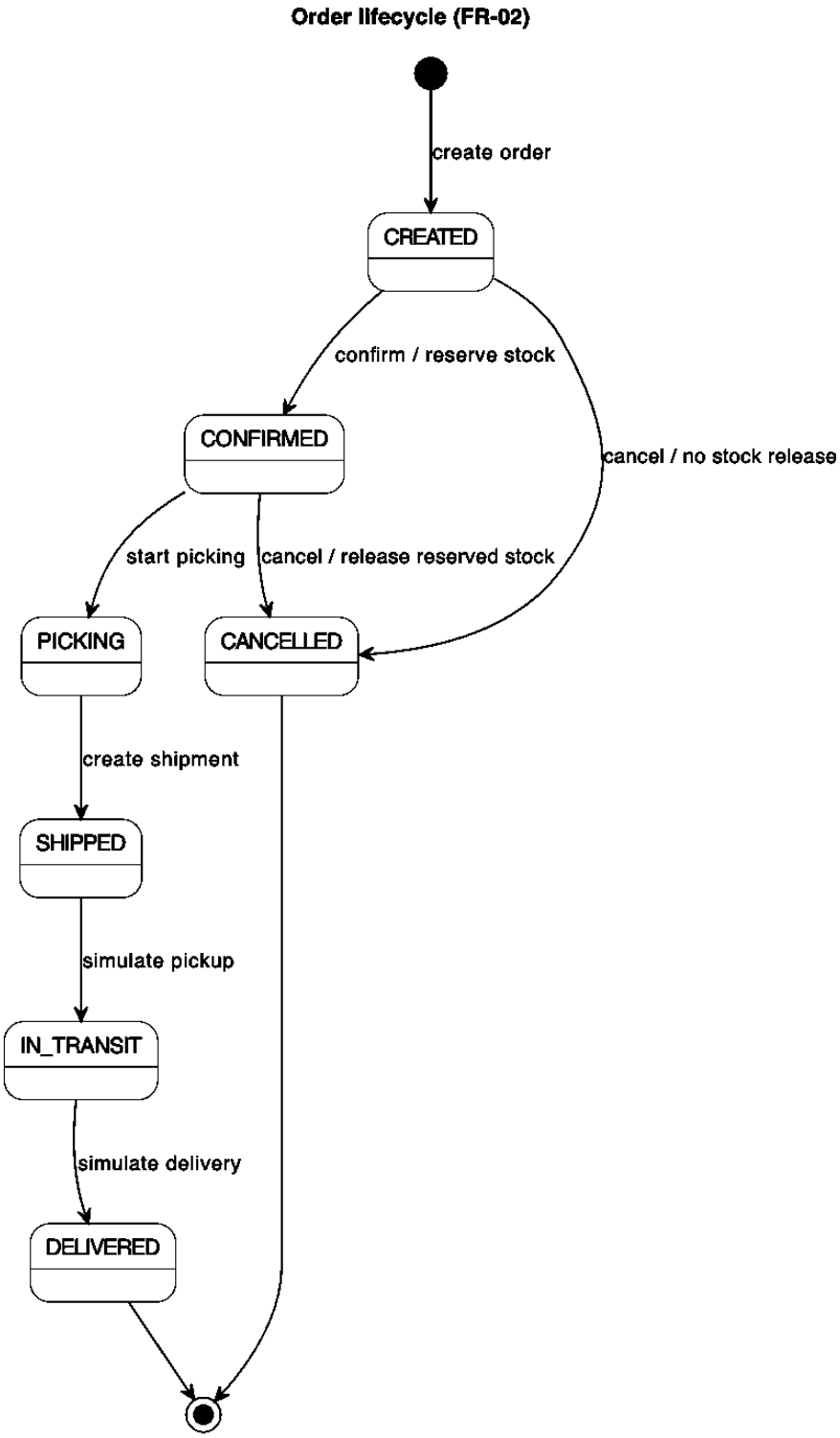


Figure 2. Order transitions and stock effects: each transition states its stock effect — reserve, release or none. Source.

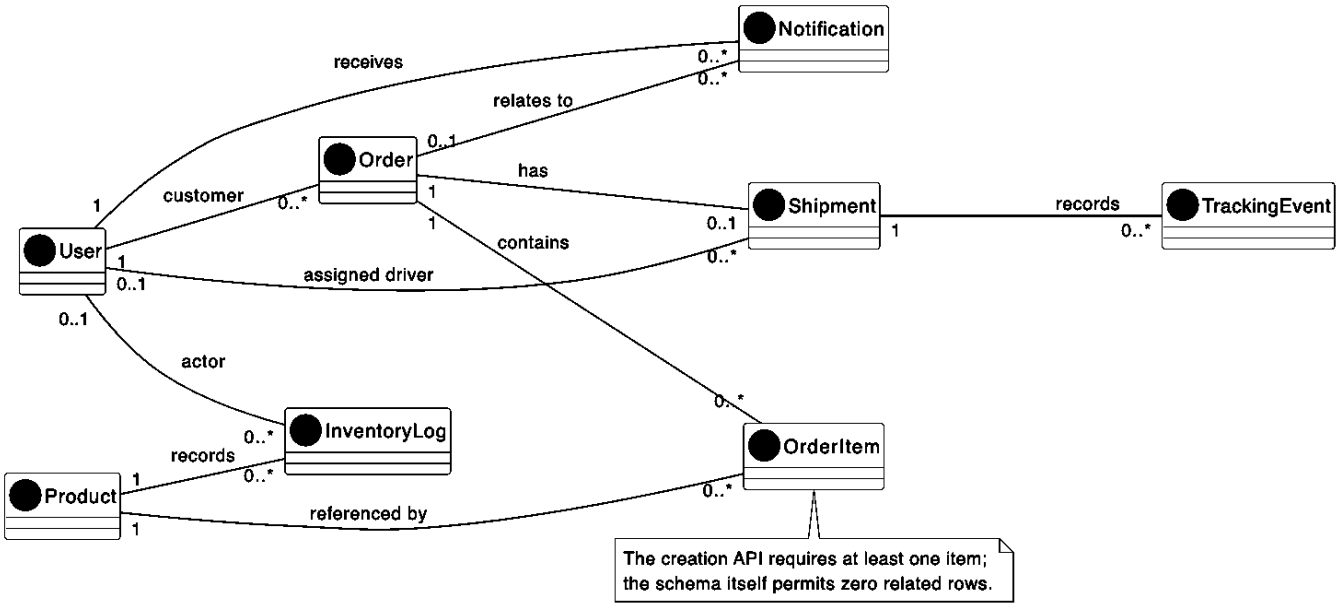


Figure 3. Data relationships: the multiplicities above, drawn as an ER model. Source.

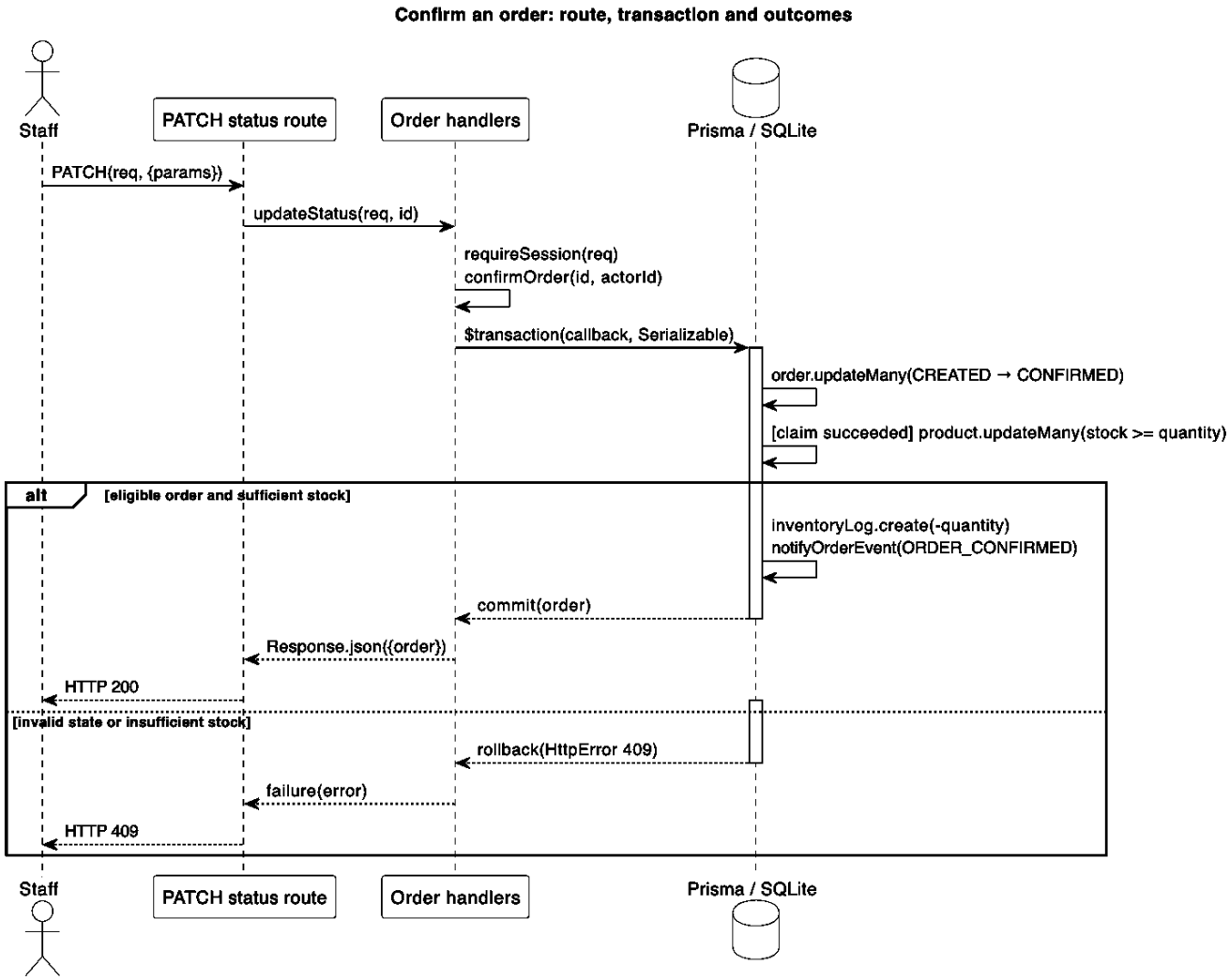


Figure 4. Confirmation transaction: session guard, conditional writes, rollback on failure or short stock. Source.

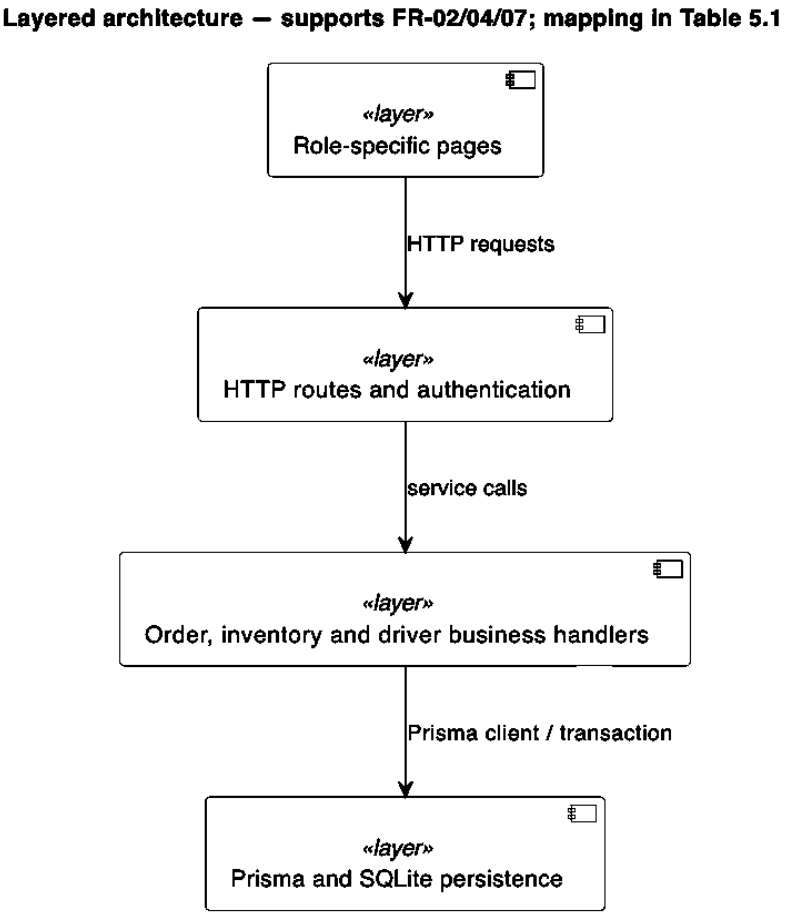


Figure 5. Four layers: role pages call HTTP routes, routes call handlers, handlers use Prisma/SQLite. Source.

5. Implementation

The confirmation transaction (order-api.ts:245–293) reserves stock. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED; CANCELLED from CREATED or CONFIRMED. Cancellations restock unless the order was CREATED. Staff cancellation is restricted to CONFIRMED.

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CREATED	rejected
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

Design	Implementation	Verification
Guards: CREATED, sufficient stock	order-api.ts:250–253, 276–280	Orders
Confirmation transaction: atomic writes; rollback on insufficient stock; repeats rejected (409)	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; kept after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04	order-api.ts:281, 398	Inventory
FR-05/06/07 handlers	notify.ts:19, reports-api.ts:105, http.ts:58 — follow the same handler pattern (separate modules)	Notifications, Reports, Auth

6. Verification

CI records 38 unit and 140 integration checks; 2 integrity checks — evidence-chain checks, not product behaviour: the pipeline refuses to run without an explicit confirmation flag (`RESET_DEMO_DATA=YES`), and the packaged bundle must match the verified source (`SHA256SUMS.txt`) (integrity checks); the integration log "tests 140 / pass 140", the unit.log "Tests 38 passed (38)".

Loopback measurement (11 Nov; 30 requests per endpoint across 6 endpoints: orders.list, orders.list.filtered, inventory.list, inventory.logs, notifications.list, reports.overview) gave API p95 3.2–13.6 ms and FCP p95 64–96 ms against 2 s and 3 s targets — loopback only, not representative of public networks or high concurrency.

The reviews also served as static verification ([Table A](§2)).

7. Evolution

Three reviews drove the changes: the 17 Oct review found the confirmation defect, fixed by the §5 transaction; the 10–11 Nov review aligned diagrams and traceability with the implementation and replaced the NFR-02 availability window with CI-run checks and scoped NFR-01 measurement to the local loopback; the 11 Nov review added the direct export path (issue #15) and two CI-health regression tests (suite 36 → 38). All four fixes remain covered by the recorded suites (§6, §8).

8. Validation

The automated walkthrough exercises eleven core operations, each at most three in-app interface interactions. "Browser acceptance passed: 26 checks" — 20 operation checks (5 operations once, 6 twice or thrice), plus 6 cross-cutting checks (mobile viewport, dark scheme, sign-out among them).

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagrams diverged from the schema	Models corrected	10–11 Nov	Correct
Stock effects were recorded under FR-04	Moved to FR-02	17 Oct	Traceal
Early register rows had thin triggers and outcomes	Register rewritten 10–11 Nov; earlier rows in git history	10–11 Nov	Registe
The filter-then-export path took four interactions	Combined path not adopted; a direct selected-date export was added (three in-app interface interactions); issue closed	11 Nov review	Issue #
No regression test for CI health	Reached 38 tests	11 Nov	Log
NFR-02 originally set 08:00–24:00 daily availability	Replaced by CI-run checks (§3) — with local-only deployment (§1) there is no hosted environment to measure a daily window; each green run verifies the integration points per commit	10–11 Nov	Require
NFR-01 assumed a hosted measurement model	Scoped to loopback measurement; budgets unchanged (§6)	10–11 Nov review	measur

10. Evidence

Baseline record — 38 unit, 140 integration, 2 integrity from the 12 Nov regression run; 26 browser checks (baseline).

Traceability.